

“CLICKED ON AD”

USING PYTHON & MACHINE LEARNING

LIWU
DEC 21, 2024

AGENDA



1. Introduction & Business Problem
2. Data Description
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Model Development & Comparison
6. Business Recommendations

INTRODUCTION & BUSINESS PROBLEM



“How can we predict if a user will **click on an advertisement?**”

- **Boosts CTR (Click-Through Rate):** Encourages more user interactions with ads.
- **Enhances Targeting:** Focuses on likely-to-engage users.
- **Actionable Insights:** Refines marketing strategies effectively.
- **Increases ROI (Return on Investment) :** Maximizes profit from ad spend.
- **Better User Experience:** Shows relevant ads to users.

DATA DESCRIPTION



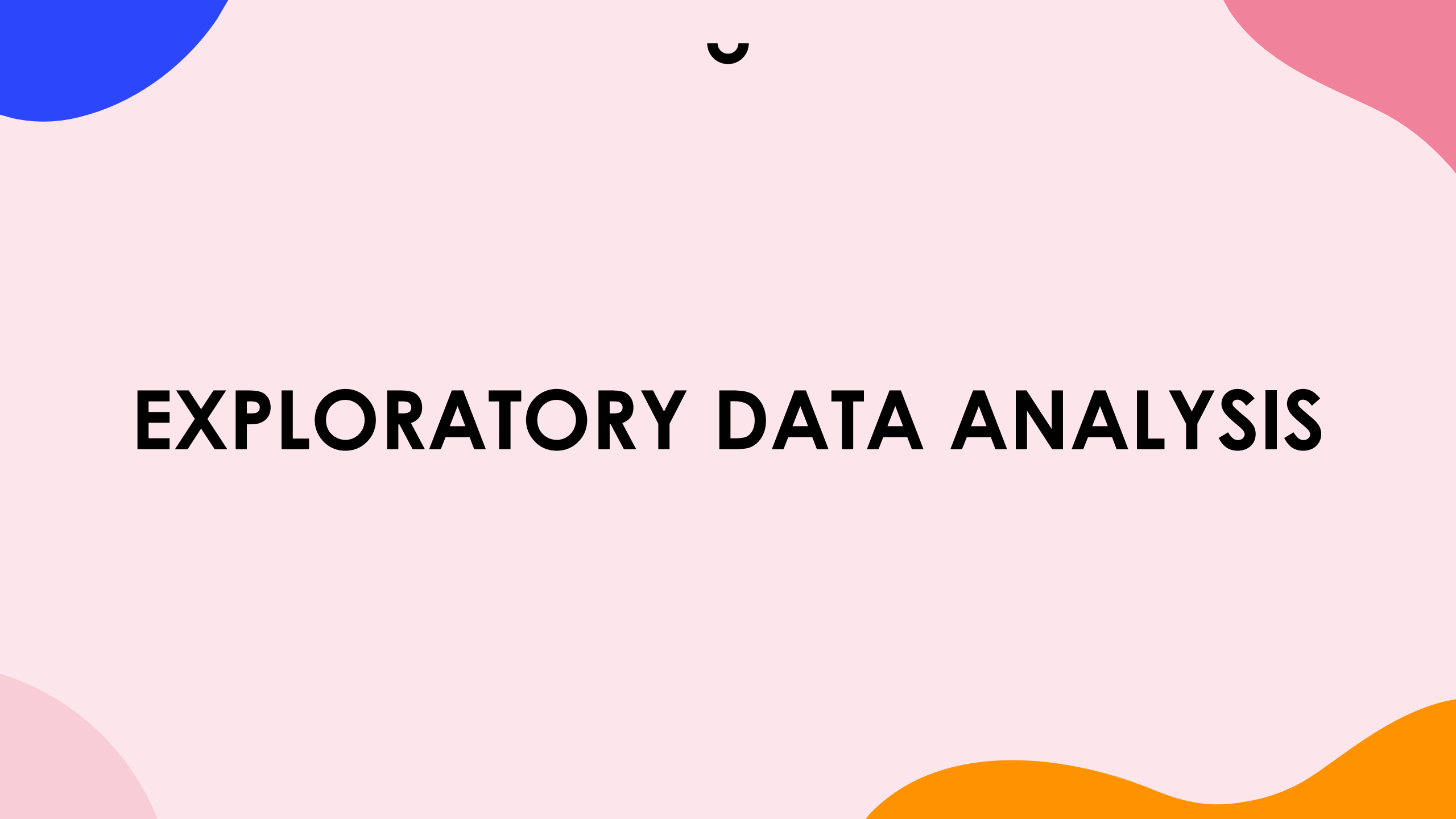
	Daily Time Spent on Site	Age	Area Income	Daily Internet Usage	Ad Topic Line	City	Gender	Country	Timestamp	Clicked on Ad
0	49.21	30.0	NaN	201.58	Total 5thgeneration standardization	West Richard	Female	Qatar	2016-07-21 10:54:35	1
1	51.65	51.0	74535.94	188.56	Function-based incremental standardization	Edwardmouth	Female	Cameroon	2016-01-31 05:12:44	0
2	47.64	29.0	53431.35	200.71	Digitized content-based circuit	East Carlos	Female	Burundi	2016-03-09 00:41:46	0
3	55.60	24.0	52252.91	167.22	Configurable disintermediate throughput	North Daniel	Female	Guinea	2016-06-11 06:47:55	1
4	78.84	27.0	69646.35	239.32	Front-line bandwidth-monitored capacity	Johnstad	Female	Spain	2016-05-21 01:36:16	0
...	...	...	...	...	...	...	...	...	...	...
2995	44.33	36.0	61610.05	140.46	Multi-layered user-facing paradigm	Sanchezland	Female	Uganda	2016-07-01 01:12:04	1
2996	80.03	42.0	60514.05	243.61	Innovative interactive portal	Lake Jennifer	Female	Czech Republic	2016-04-06 11:24:21	1
2997	41.73	31.0	61840.26	207.17	Profound executive flexibility	West Angelabury	Male	Singapore	2016-01-03 03:22:15	1
2998	41.73	28.0	51501.38	120.49	Managed zero tolerance concept	Kennedyfurt	Male	Luxembourg	2016-05-28 12:20:15	0
2999	46.61	50.0	43974.49	123.13	Realigned content-based leverage	North Samantha	Female	Malawi	2016-04-04 07:07:46	1

DATA DESCRIPTION

- 2016 **Jan-July**
- **3,000** observations
- Duplicates: **31**
- Missing Values:

	Number of Missing	Percentage(%)
City	33	1.10
Area Income	31	1.03
Country	28	0.93
Age	24	0.80
Gender	23	0.77

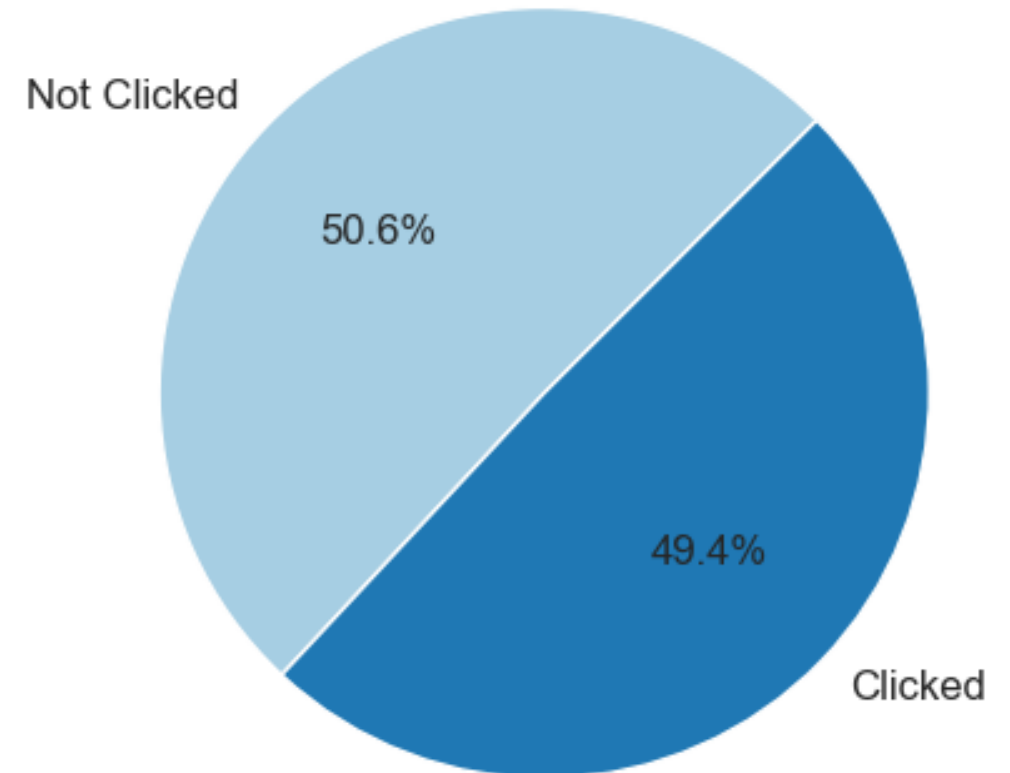
Target Variable	Clicked on Ad
Numerical Variables	• Timestamp (Datetime) • Daily Time Spend on Site • Daily Internet Usage • Age • Area Income
Categorical Variables	• Gender • Ad Topic Line • City • Country



EXPLORATORY DATA ANALYSIS

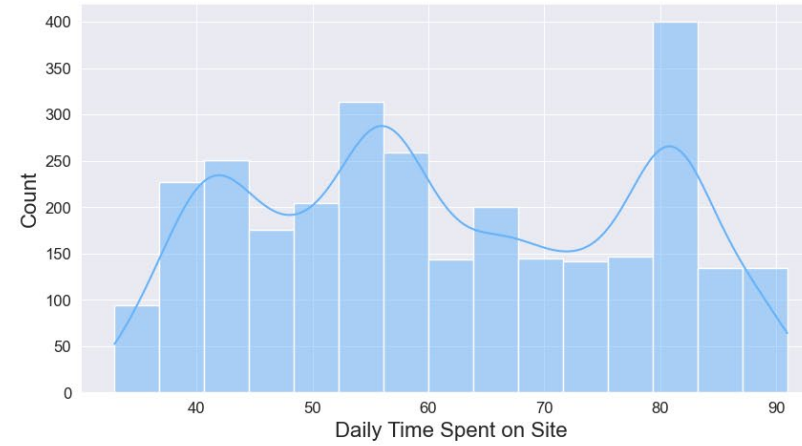
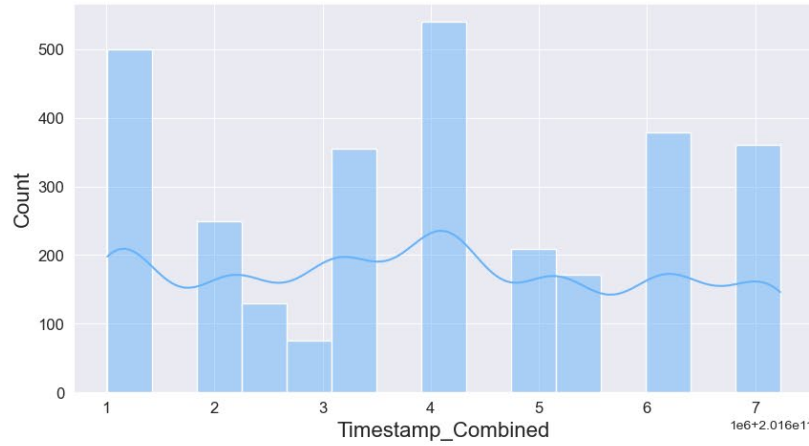


“CLICKED ON AD”



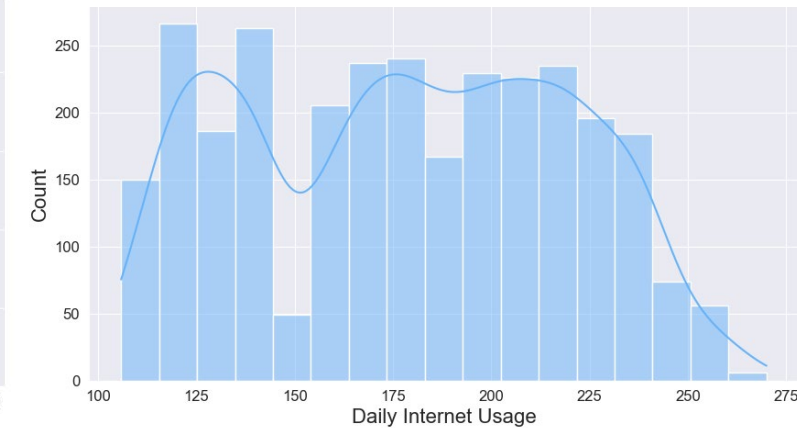
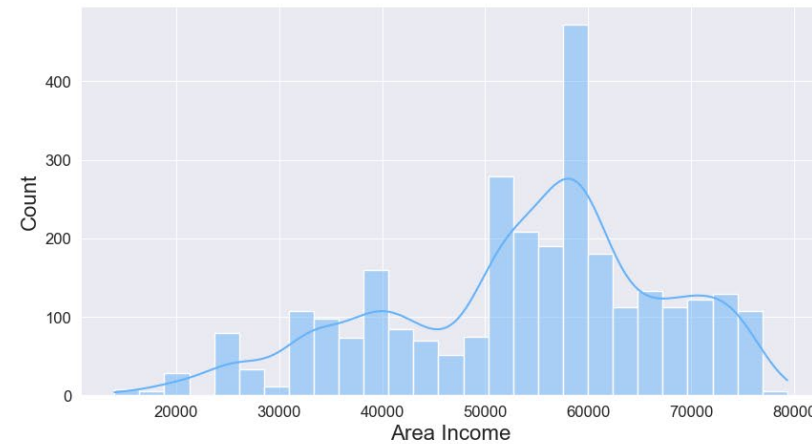
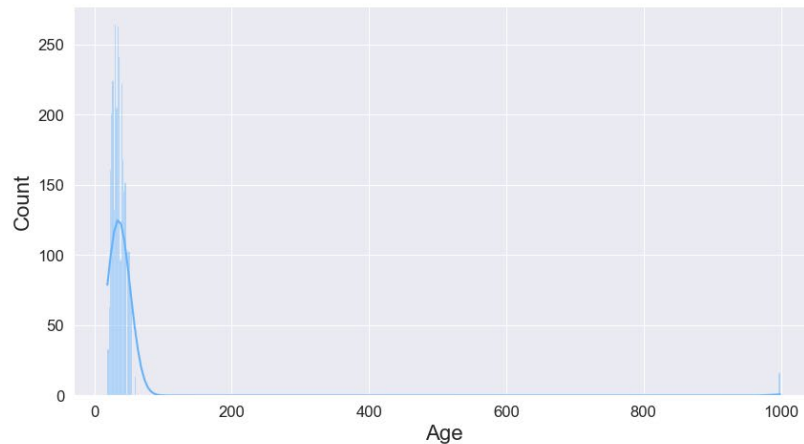


NUMERICAL VARIABLES

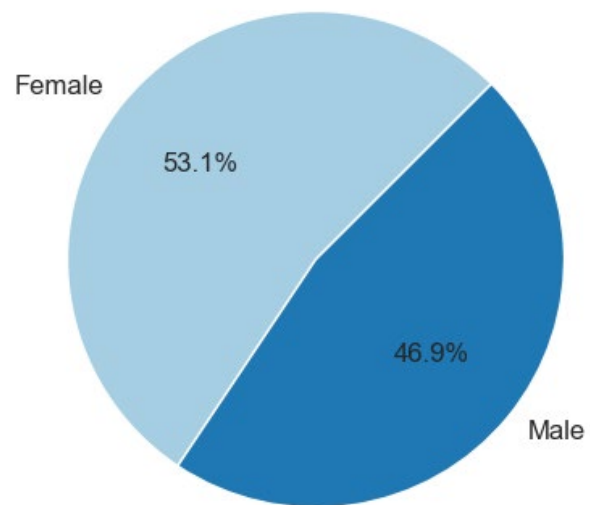


SKEWNESS

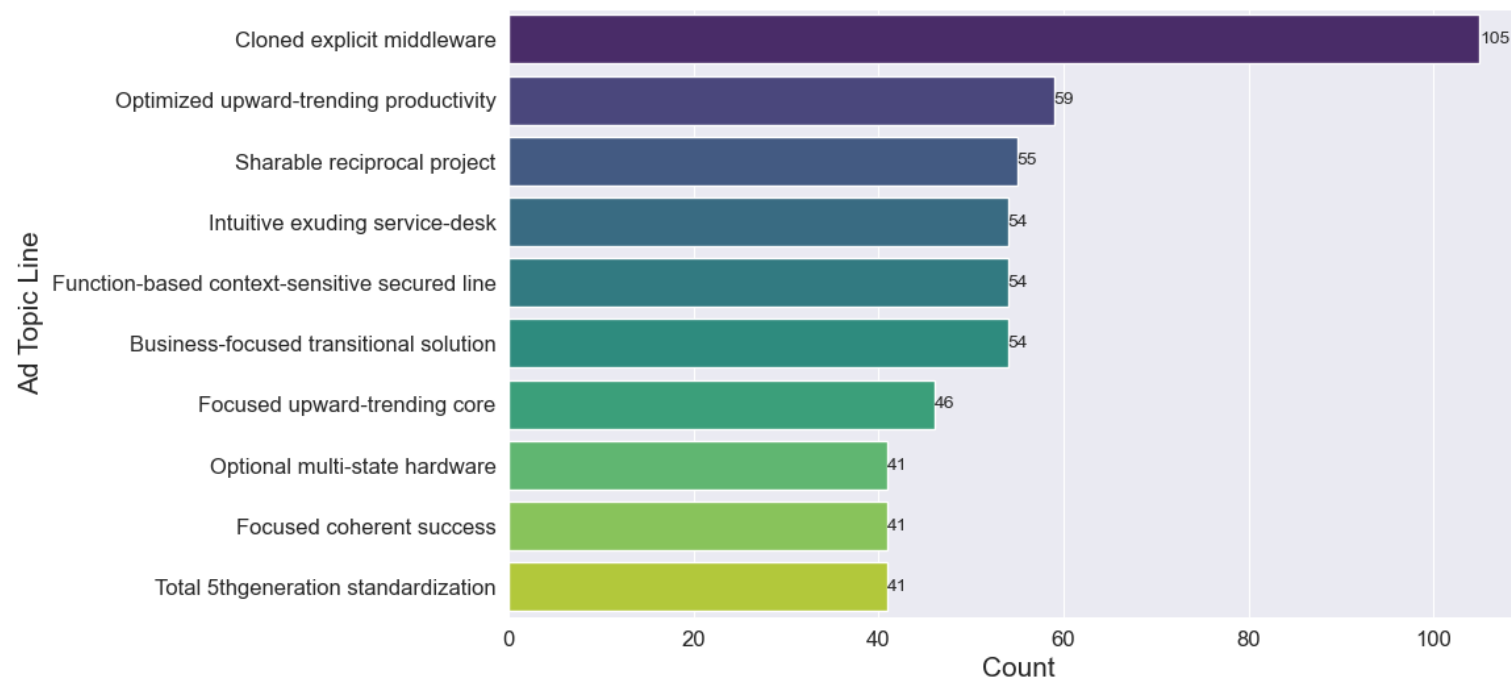
- Timestamp_Combined: **0.06**
- Daily Time Spend on Site: **0.09**
- Age: **13.17**
- Area Income: **-0.51**
- Daily Internet Usage: **-0.01**



CATEGORICAL VARIABLES

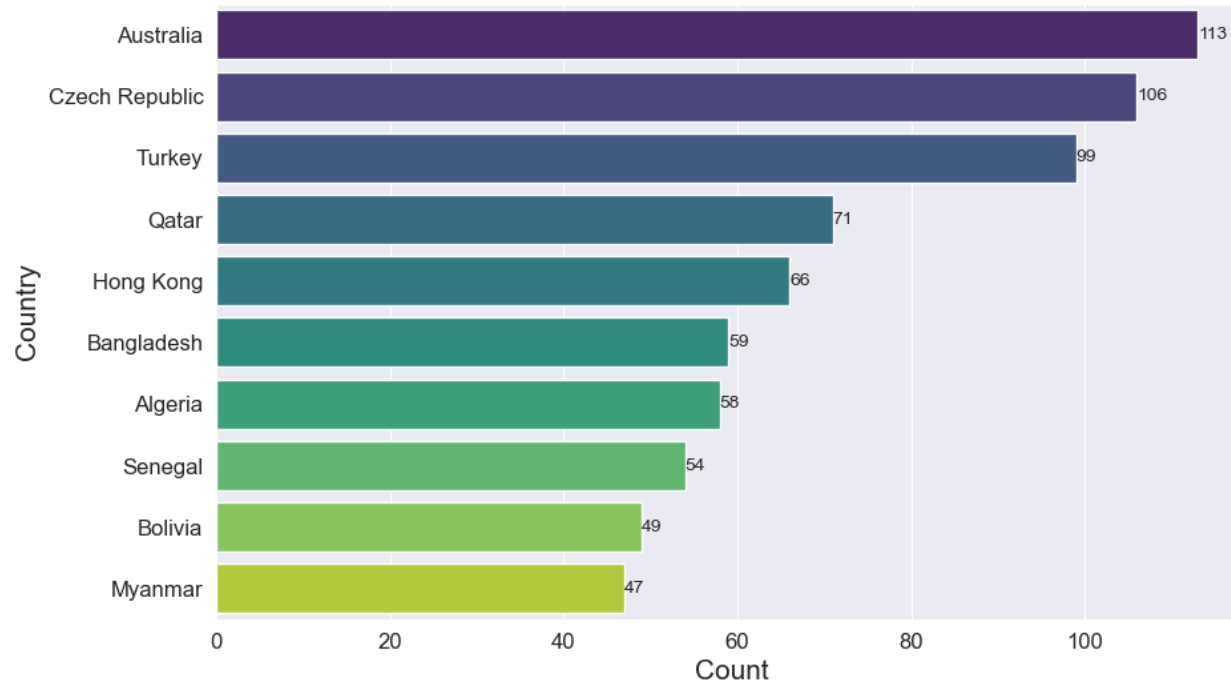


Gender

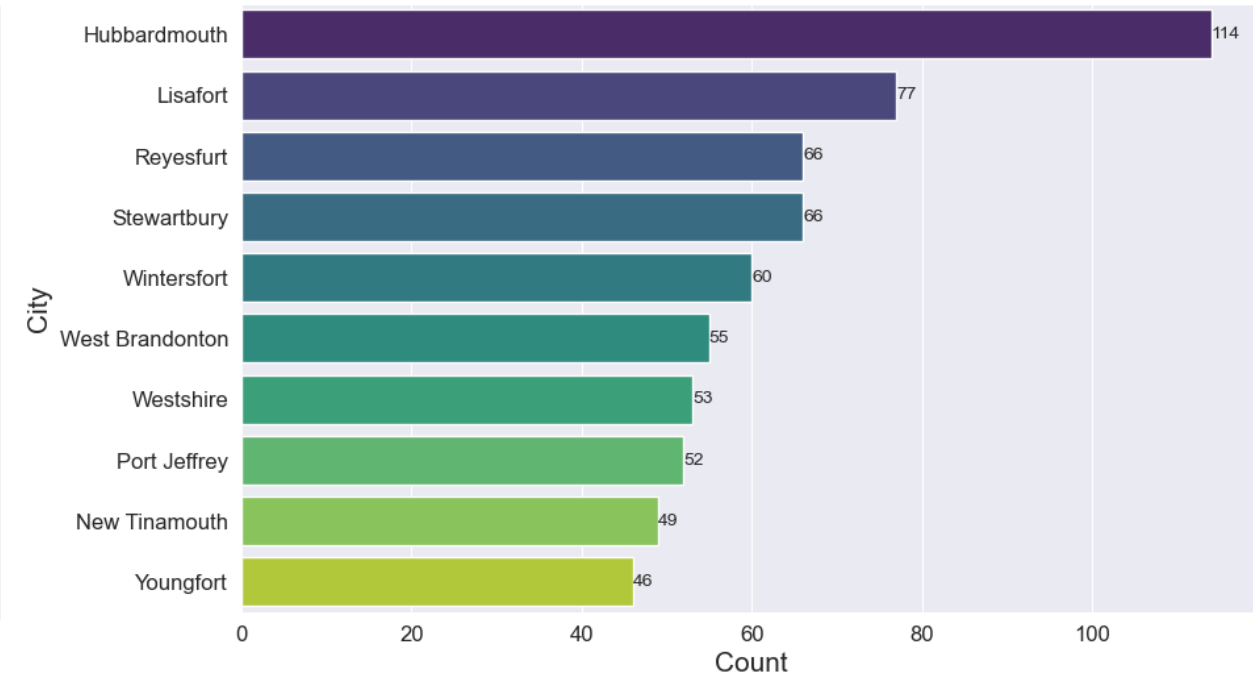


Ad Topic Line (417 levels)

CATEGORICAL VARIABLES



Country (186 levels)



City (407 levels)

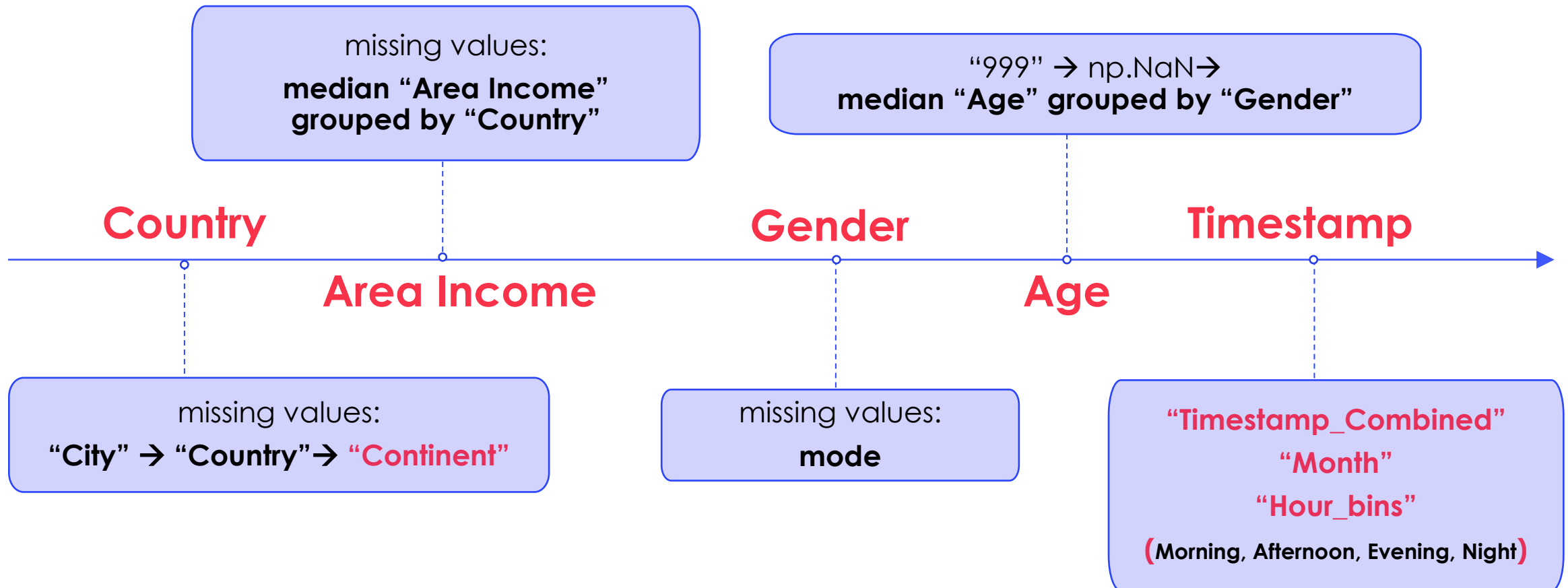


FEATURE ENGINEERING



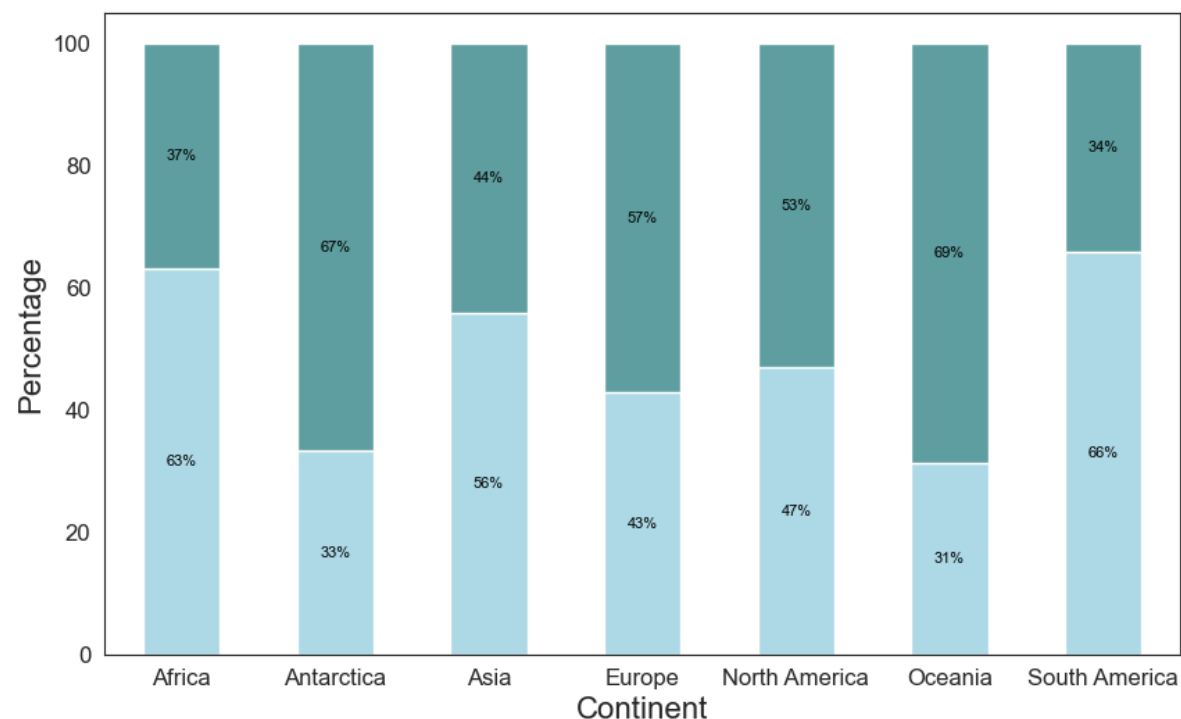
KEY STEPS

- Handle missing values
- Handle Outliers
- **Feature Creation** (Datetime: Month, Hour)
- **Feature Transformation** (Segmentation)
- **Feature Selection**
 - Statistical Tests (Chi-Square, T-test etc.)
 - Correlation, Multicollinearity (VIF)
- **One-Hot Encoding** for nominal categories (unordered)
- **Feature Scaling and Normalization** (Linear Models)

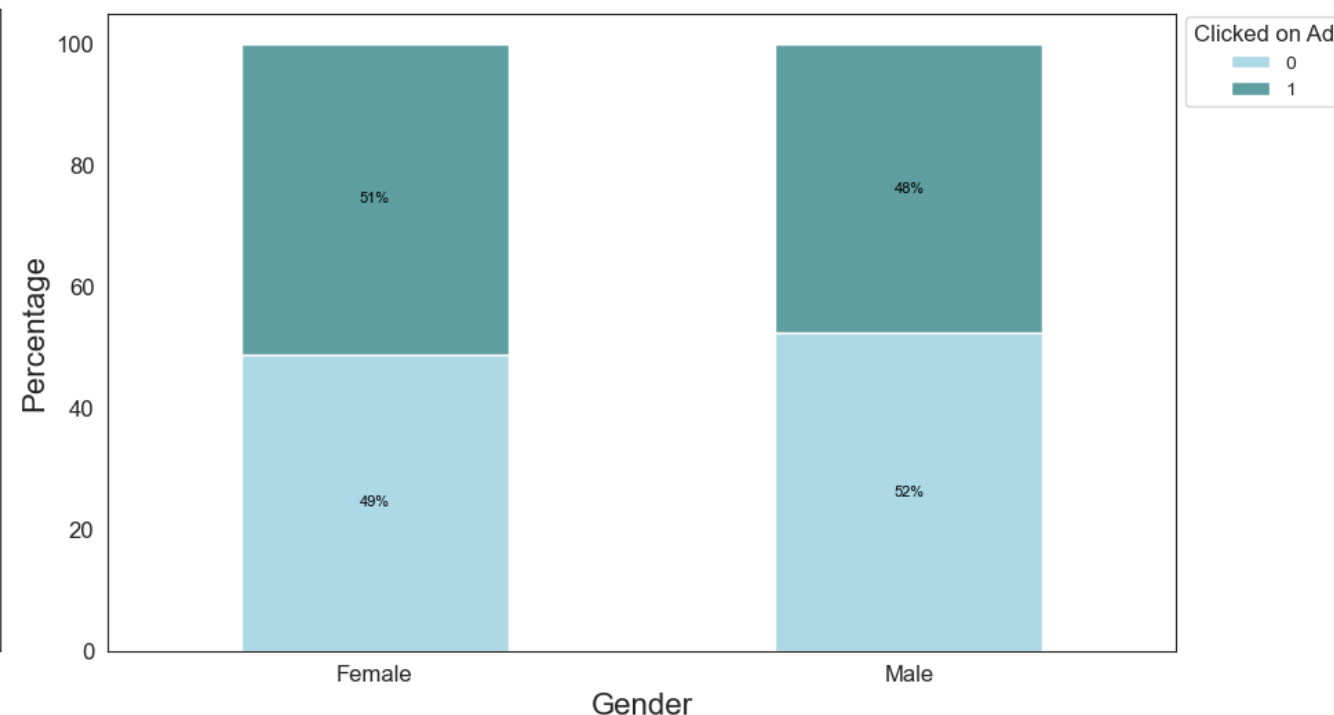


“CLICKED ON AD” VS CATEGORICAL VARIABLES

Chi-Square test



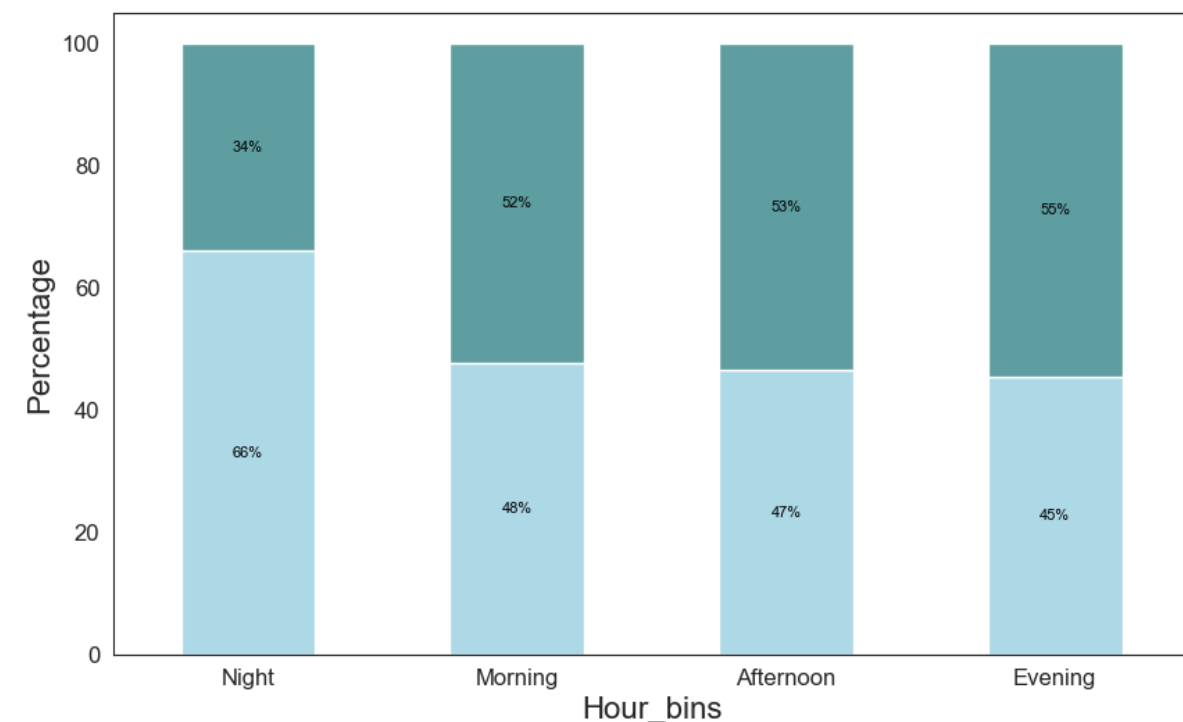
Continent - Association



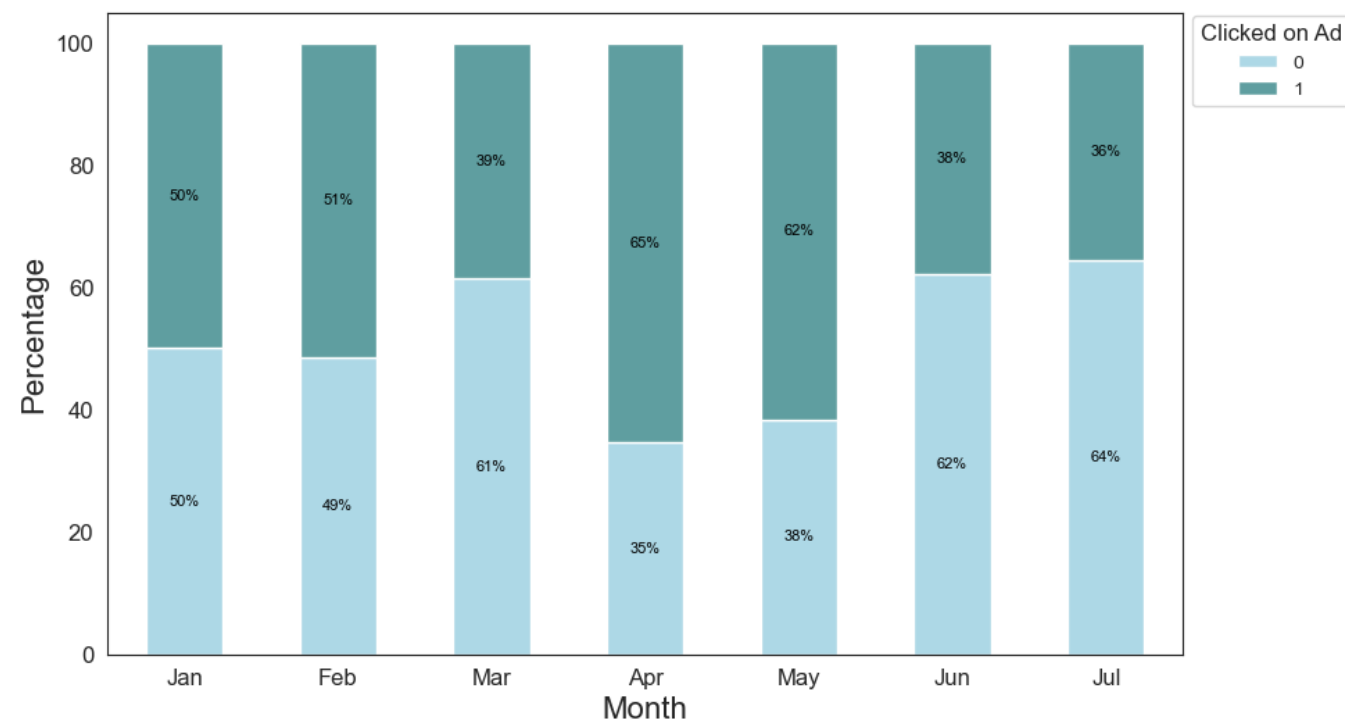
Gender - No Association

“CLICKED ON AD” VS CATEGORICAL VARIABLES

Chi-Square test



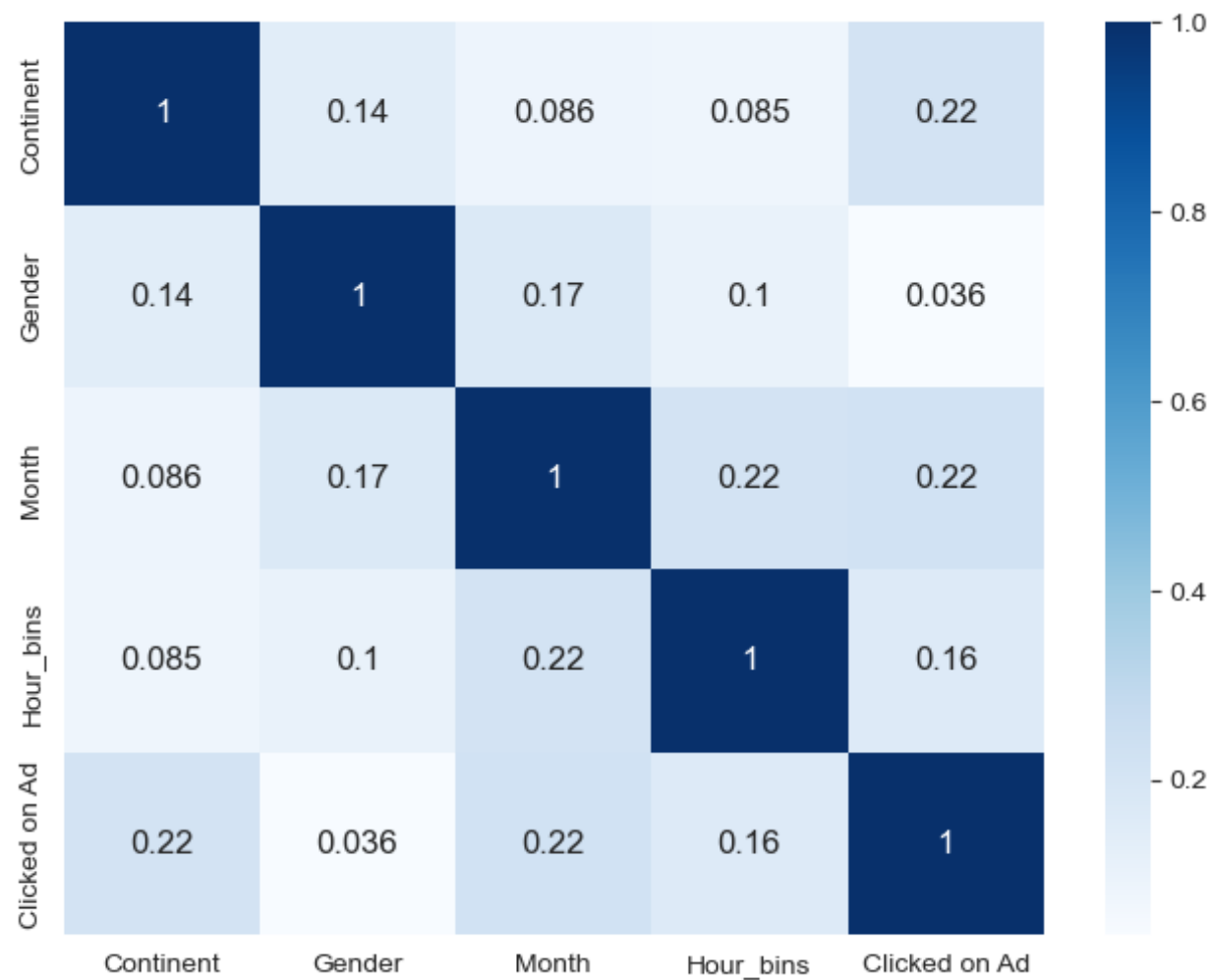
Hour_bins - Association



Month - Association

“CLICKED ON AD” VS CATEGORICAL VARIABLES

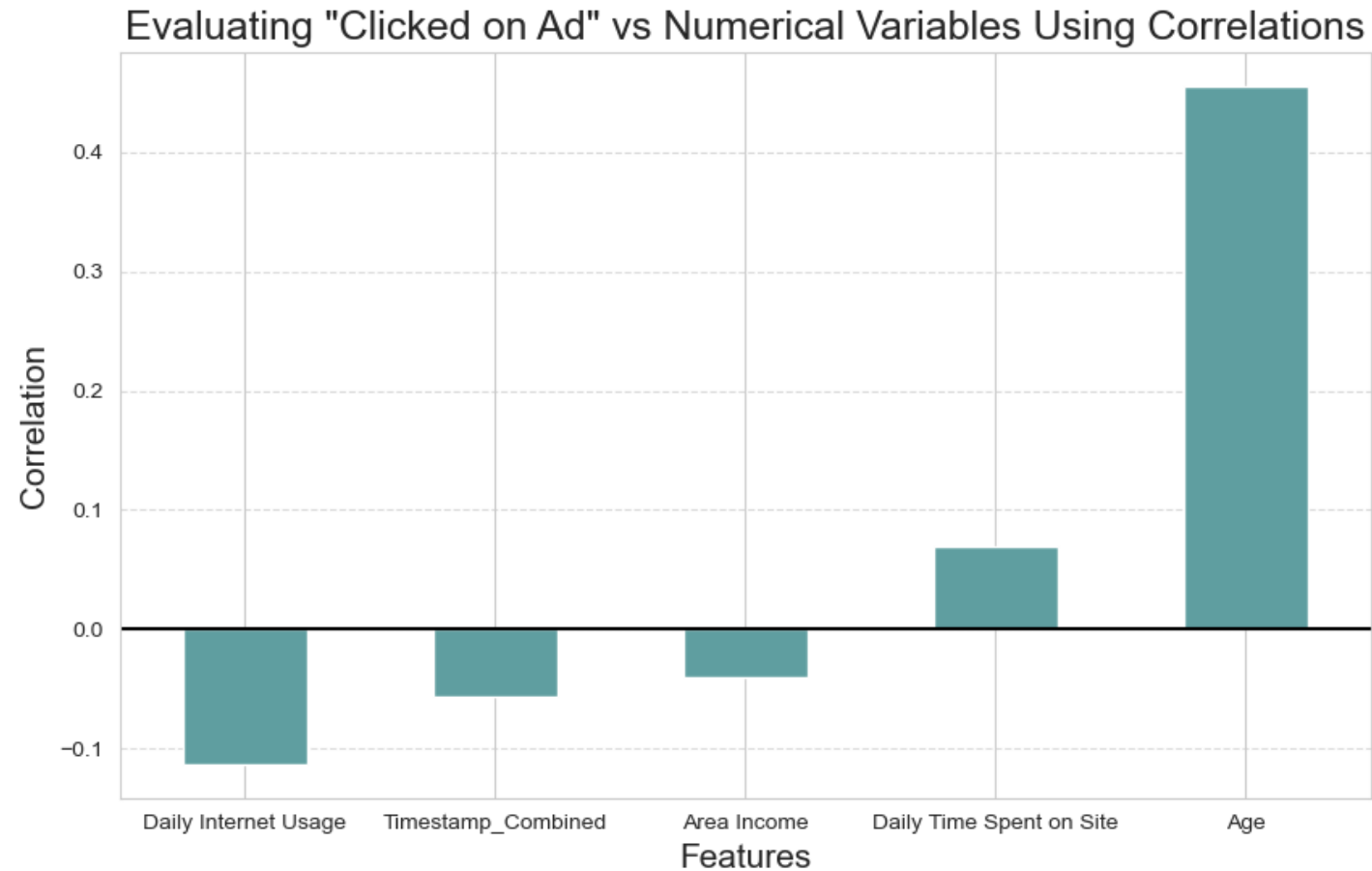
Cramer's V



"CLICKED ON AD" VS NUMERICAL VARIABLES

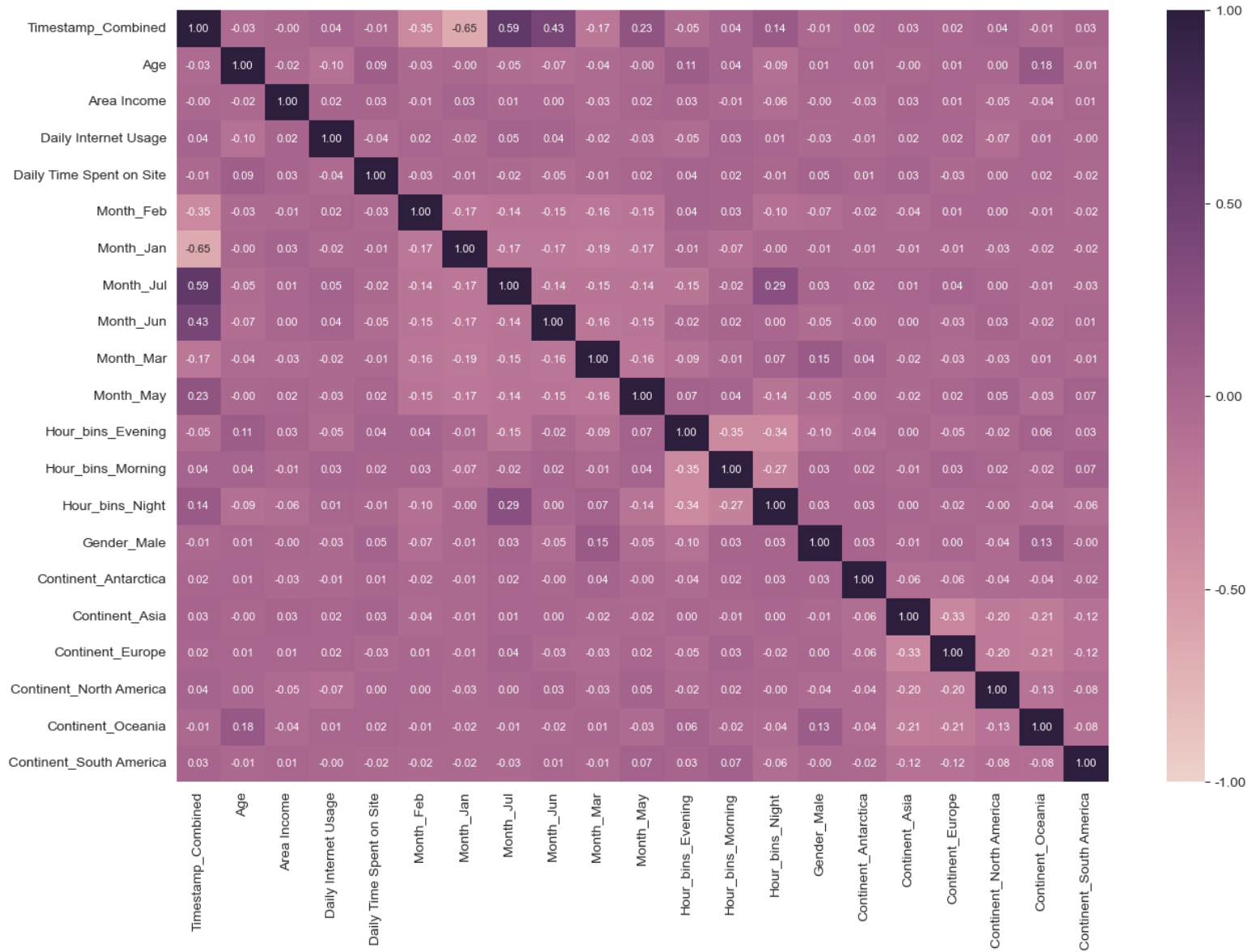


T-test



"Age" : Strongest Positive Correlation

COLLINEARITY



MULTICOLLINEARITY

Linear Model

- Remove Features (VIF > 10)
 - Unstable Coefficients
 - Reduced Interpretability
- Scale the data

Tree-Based Models

- Keep All Features
- Scale not necessary

Timestamp_Combined
(VIF 85.83)



Age
(VIF 15.48)



Daily Internet Usage
(VIF 14.42)



Area Income
(VIF 11.35)

Feature	VIF (<=10)
Daily Time Spent on Site	8.04
Month_Feb	1.49
Month_Jan	1.66
Month_Jul	1.65
Month_Jun	1.49
Month_Mar	1.64
Month_May	1.54
Hour_bins_Evening	2.05
Hour_bins_Morning	1.77
Hour_bins_Night	1.86
Gender_Male	1.96
Continent_Antarctica	1.05
Continent_Asia	1.95
Continent_Europe	1.90
Continent_North America	1.41
Continent_Oceania	1.50
Continent_South America	1.19



MODEL DEVELOPMENT

Logistic Regression

Decision Tree

Random Forest

XGBoost



ALL MODELS

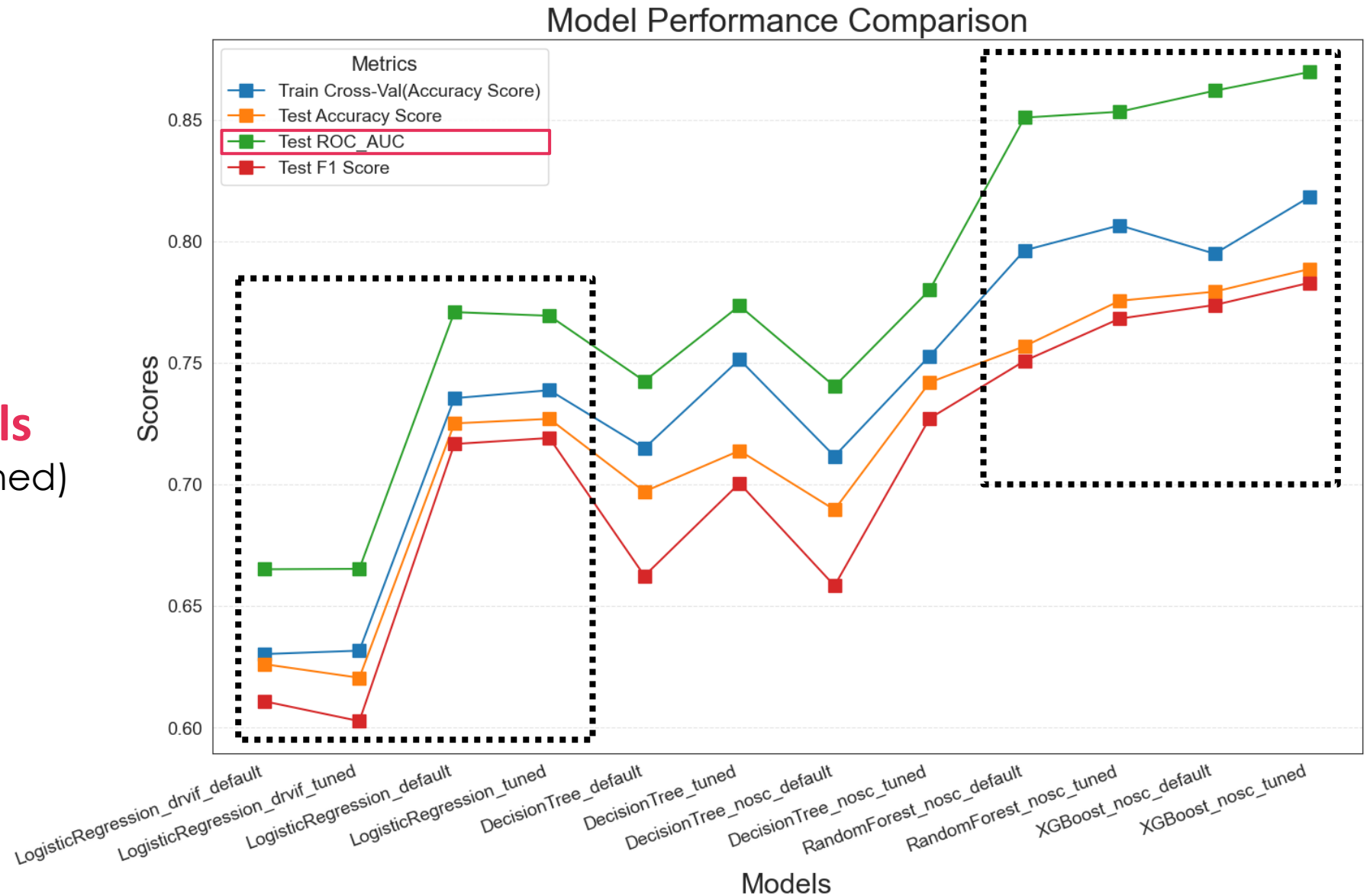
Model Selection

Linear Model

- Logistic Regression (Baseline)

Tree-Based Models

- Random Forest (Tuned)
- XGBoost (Tuned)



ALL MODELS



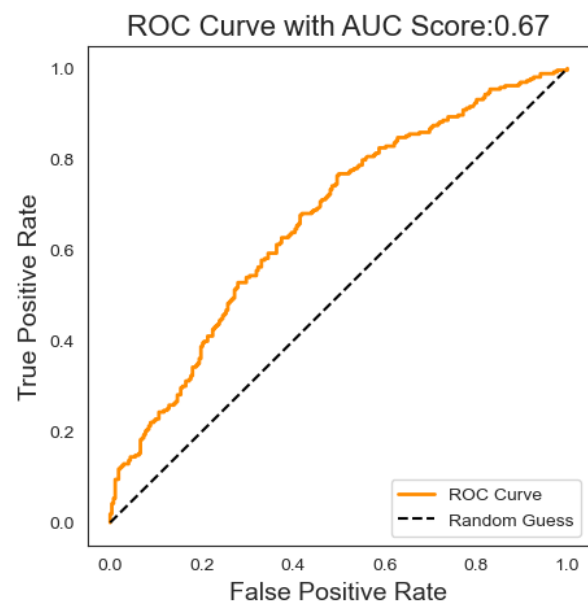
	Model	Train Cross-Val(Accuracy Score)	Test Accuracy Score	Train Cross-Val(ROC_AUC)	Test ROC_AUC	Train Cross-Val(F1 Score)	Test F1 Score
0	LogisticRegression_drvif_default	0.630325	0.626168	0.673368	0.665218	0.615050	0.610895
1	LogisticRegression_drvif_tuned	0.631729	0.620561	0.673509	0.665399	0.617407	0.602740
2	LogisticRegression_default	0.735616	0.725234	0.801229	0.771038	0.723807	0.716763
3	LogisticRegression_tuned	0.738892	0.727103	0.801315	0.769515	0.727031	0.719231
4	DecisionTree_default	0.715027	0.697196	0.775662	0.742668	0.688912	0.662500
5	DecisionTree_tuned	0.751532	0.714019	0.800933	0.773631	0.734099	0.700587
6	DecisionTree_nosc_default	0.711751	0.689720	0.772482	0.740487	0.683956	0.658436
7	DecisionTree_nosc_tuned	0.752937	0.742056	0.800080	0.780097	0.740189	0.727273
8	RandomForest_nosc_default	0.796453	0.757009	0.875372	0.851061	0.789449	0.750958
9	RandomForest_nosc_tuned	0.806745	0.775701	0.877335	0.853479	0.800553	0.768340
10	XGBoost_nosc_default	0.795037	0.779439	0.881843	0.862125	0.792159	0.773946
11	XGBoost_nosc_tuned	0.818448	0.788785	0.892335	0.869842	0.812669	0.783109



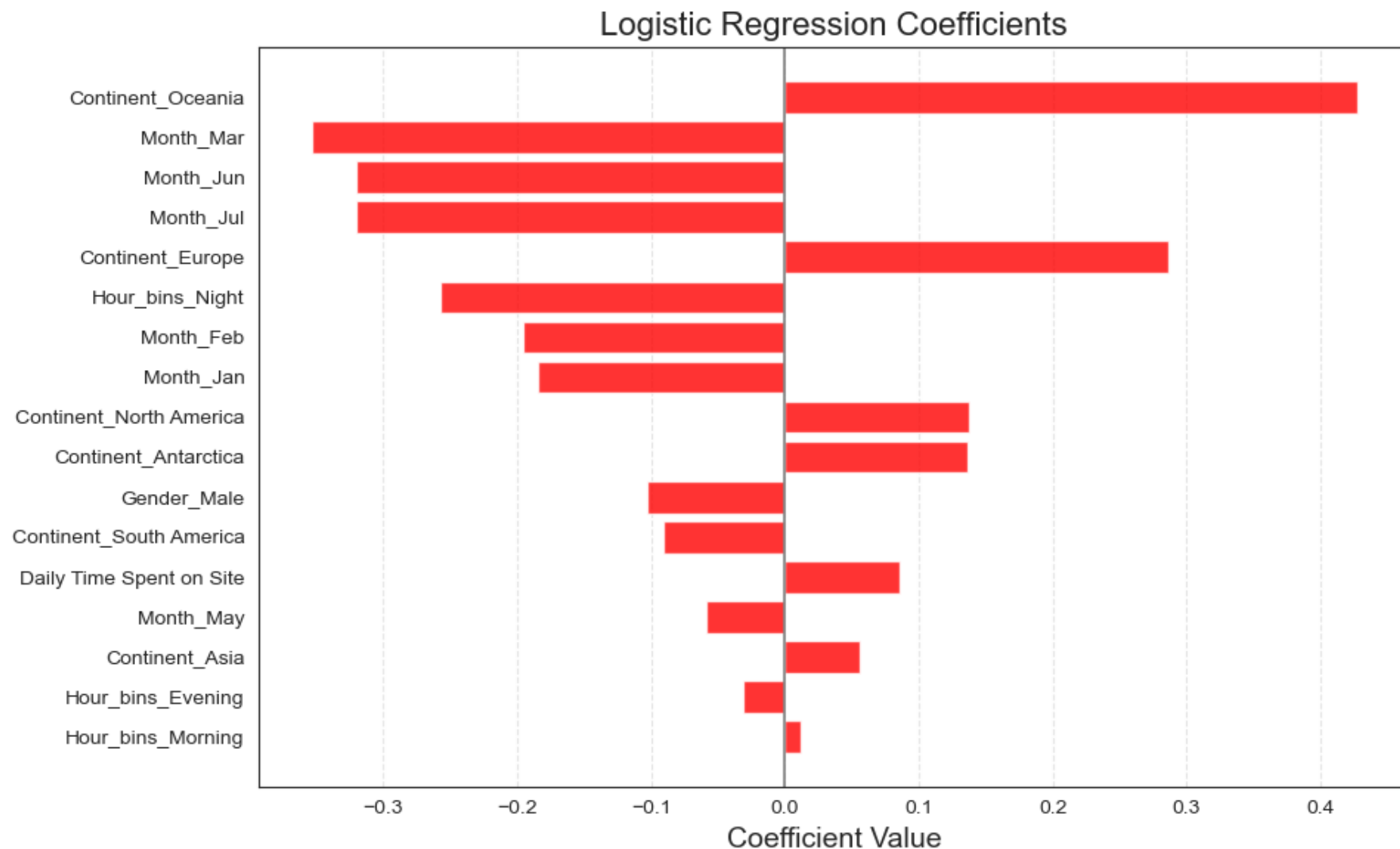
LOGISTIC REGRESSION

Features VIF $\leq$ 10

- Train Cross-Val Accuracy: **0.63**
- Test Accuracy: **0.62**
- Test ROC_AUC: **0.66**
- Test F1 Score: **0.60**



Features

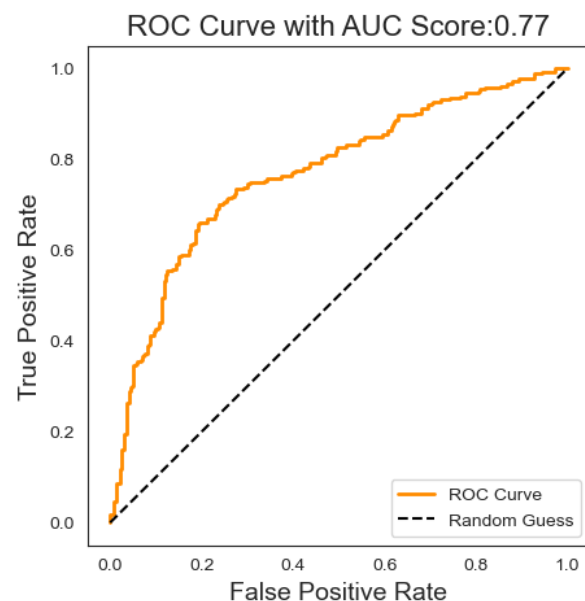




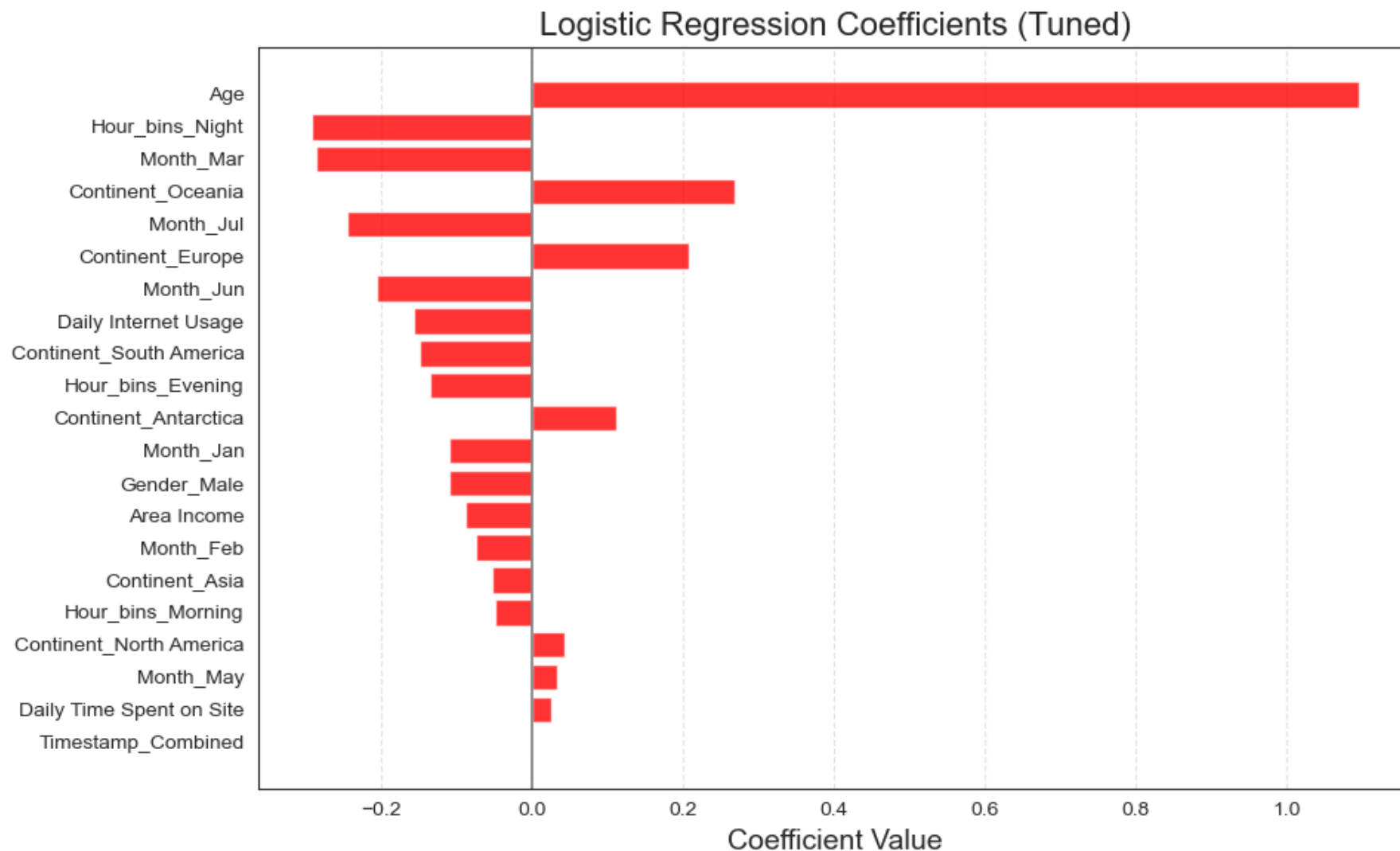
LOGISTIC REGRESSION

All Columns

- Train Cross-Val Accuracy: **0.73**
- Test Accuracy: **0.72**
- Test ROC_AUC: **0.76**
- Test F1 Score: **0.71**



Features

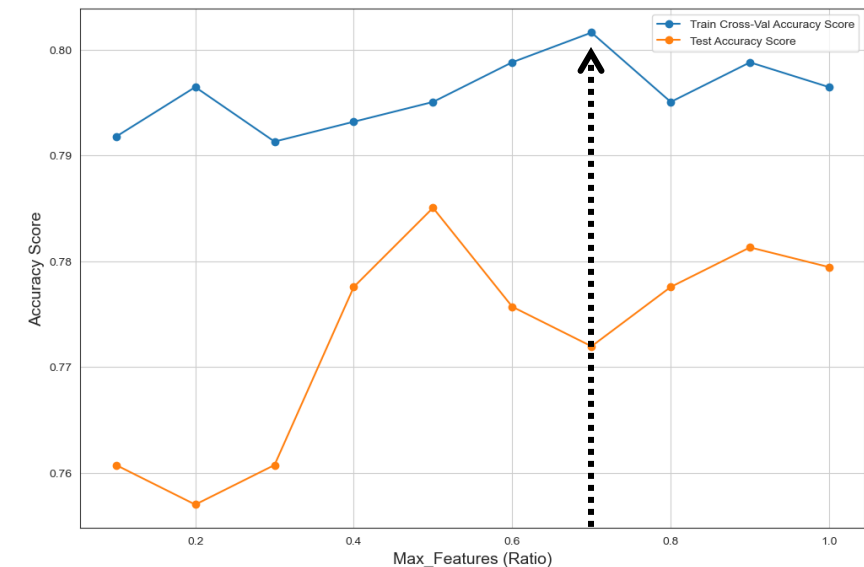
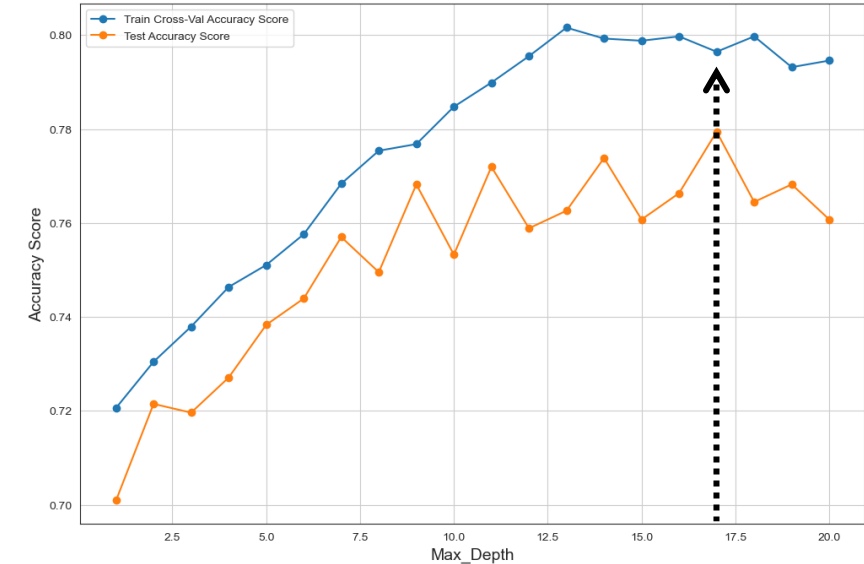
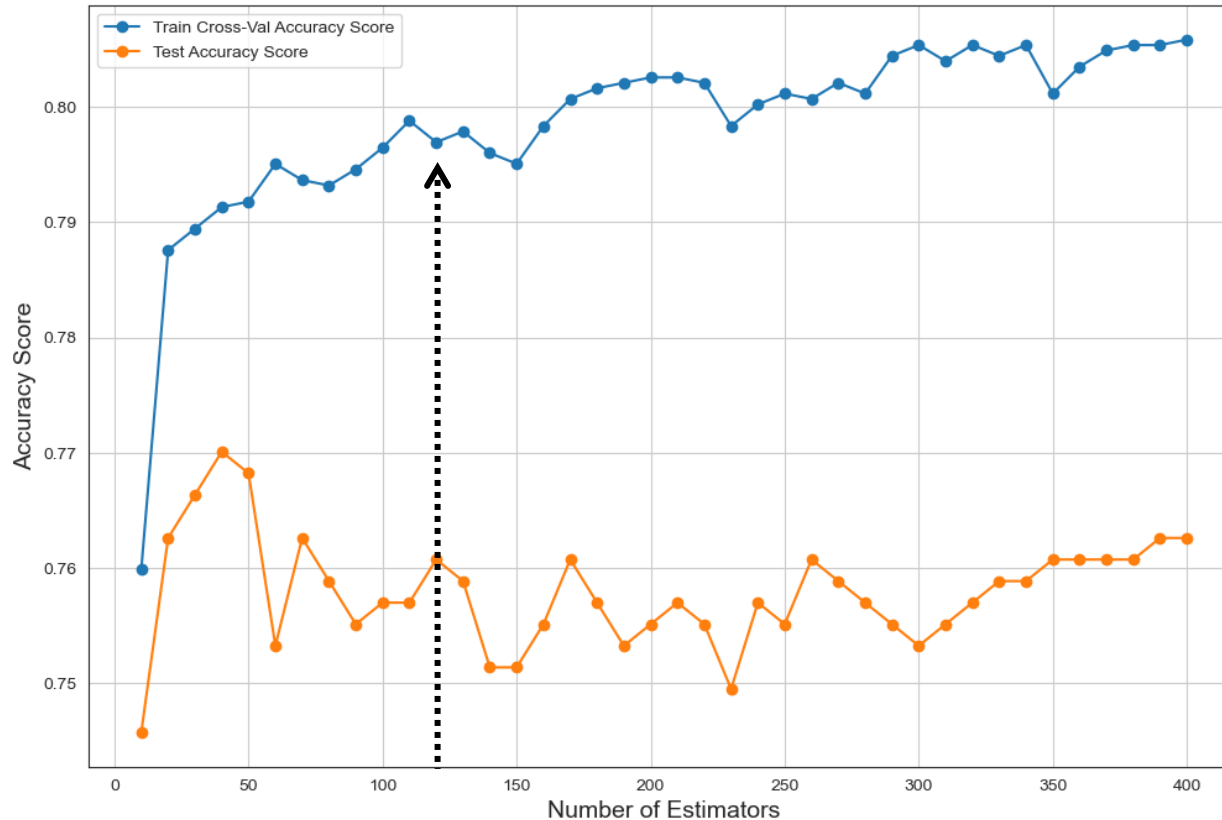




TREE-BASED MODELS

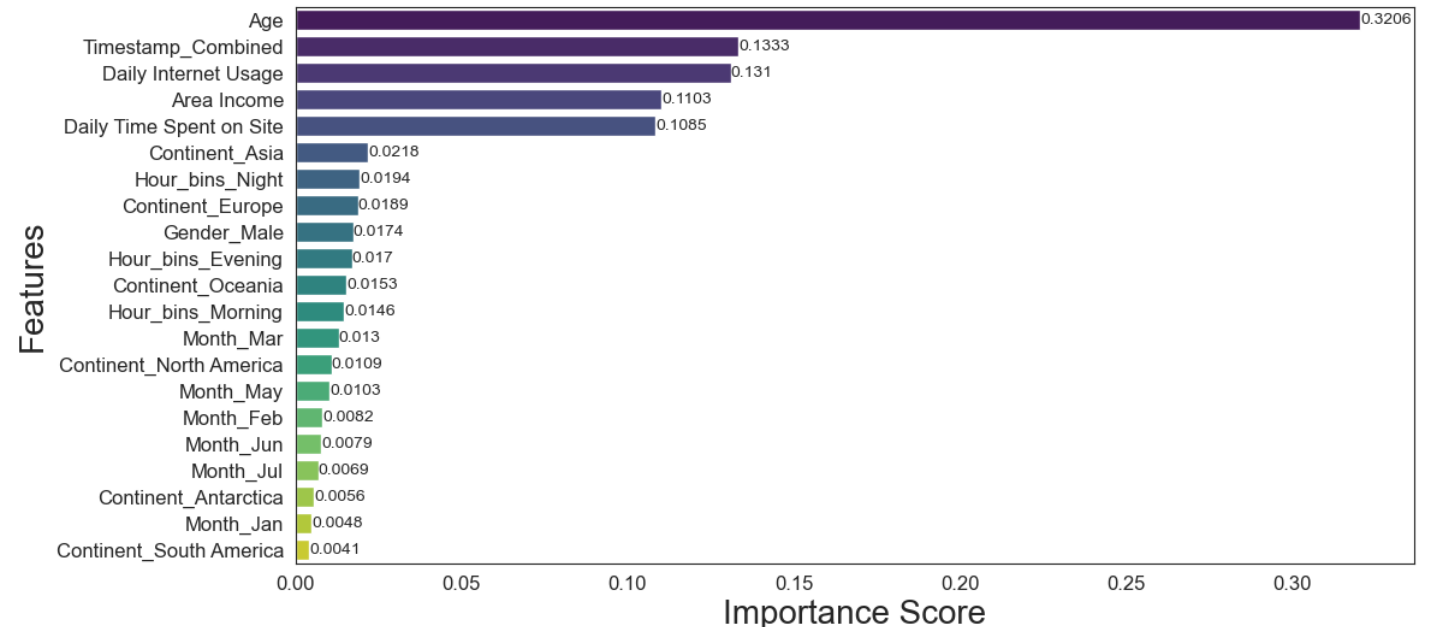
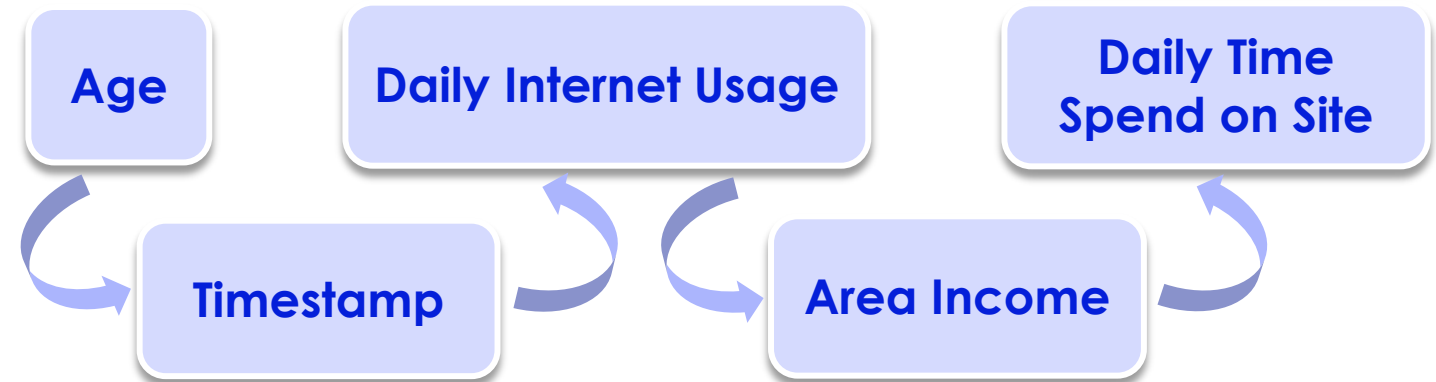
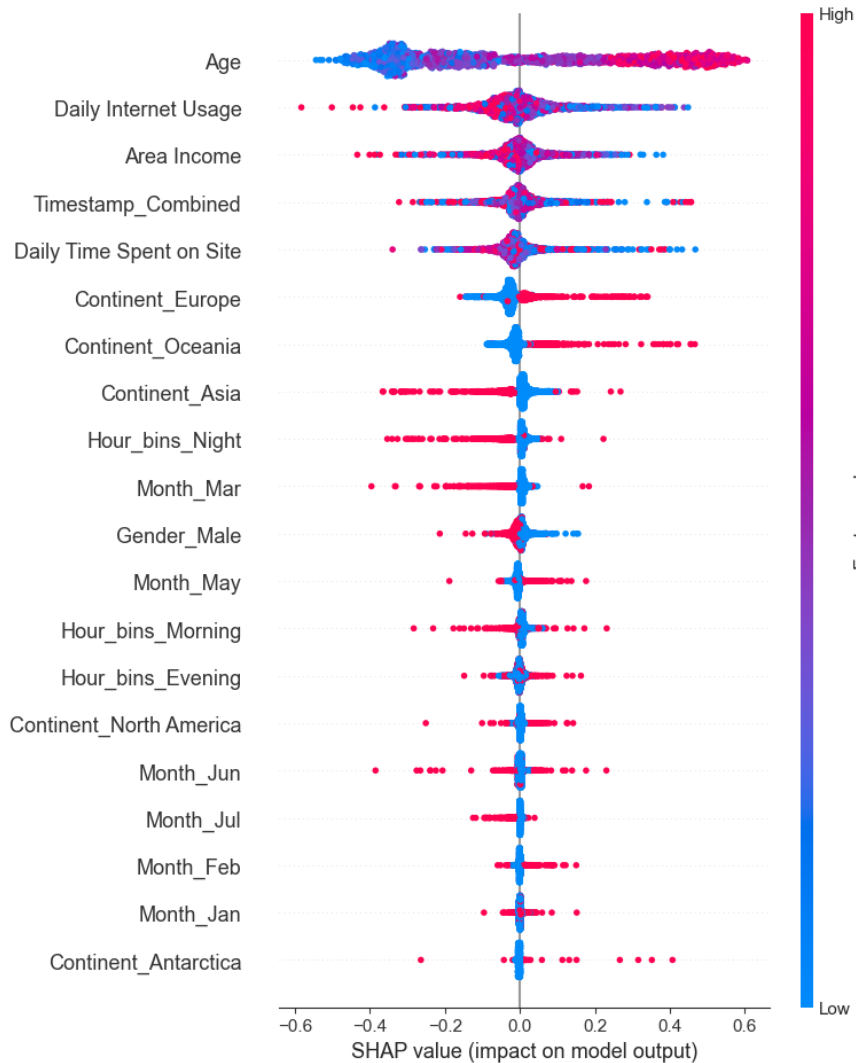
Random Forest

- Train Cross-Val Accuracy: **0.80**
 - Test Accuracy: **0.77**
 - Test ROC_AUC: **0.85**
 - Test F1 Score: **0.76**
- n_estimators: 120
 - max_depth: 17
 - max_features: 0.7



TREE-BASED MODELS

Random Forest

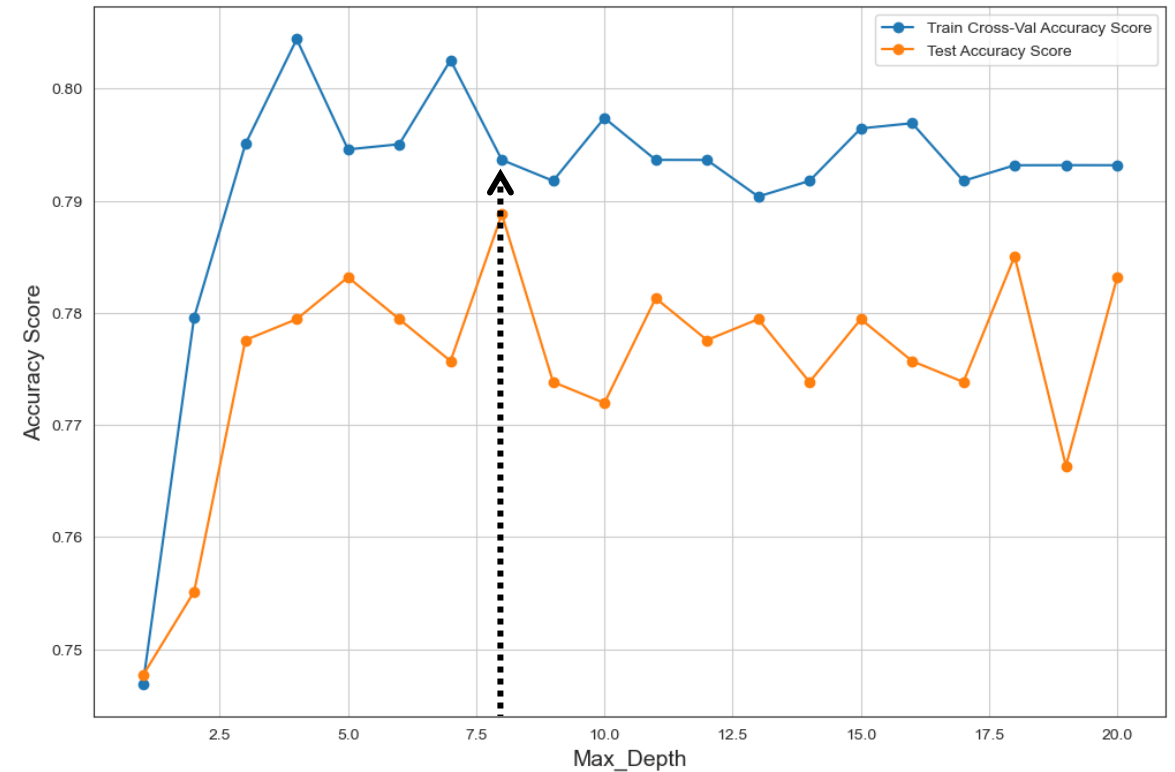
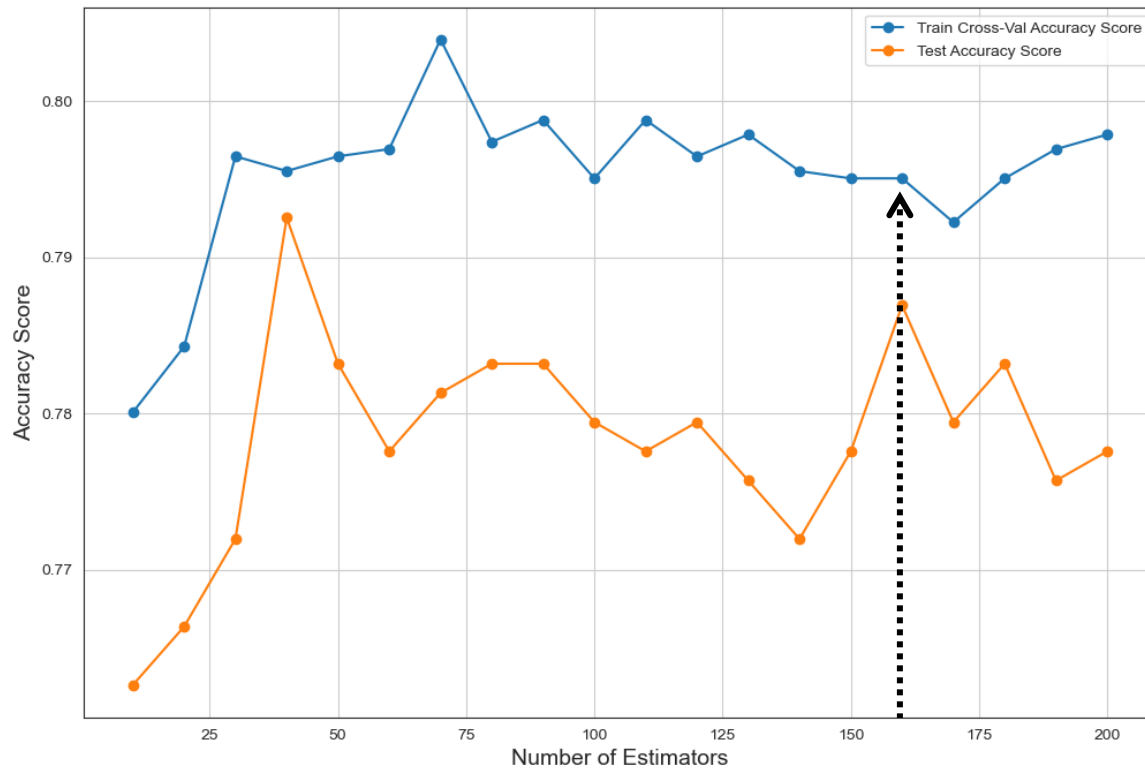




TREE-BASED MODELS

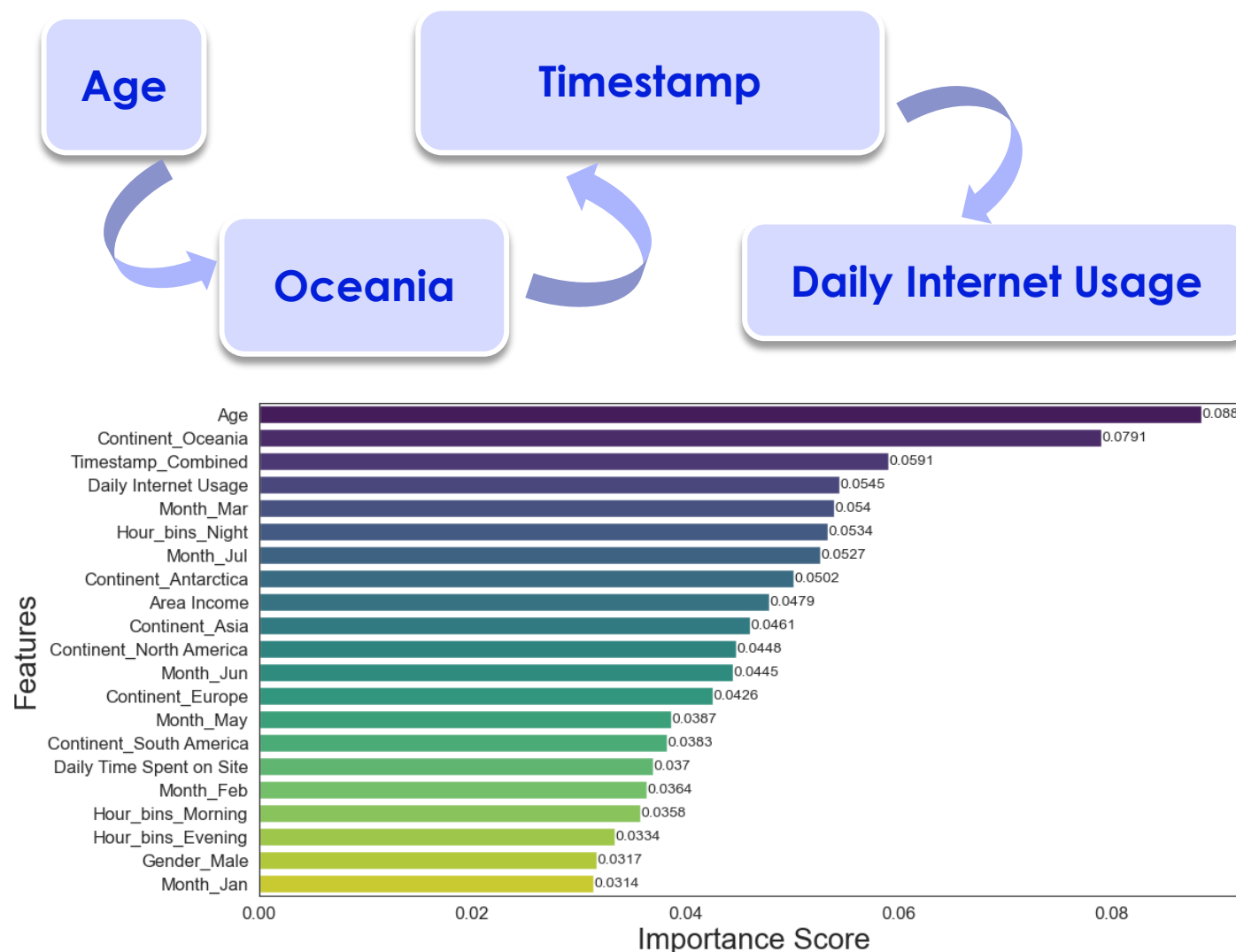
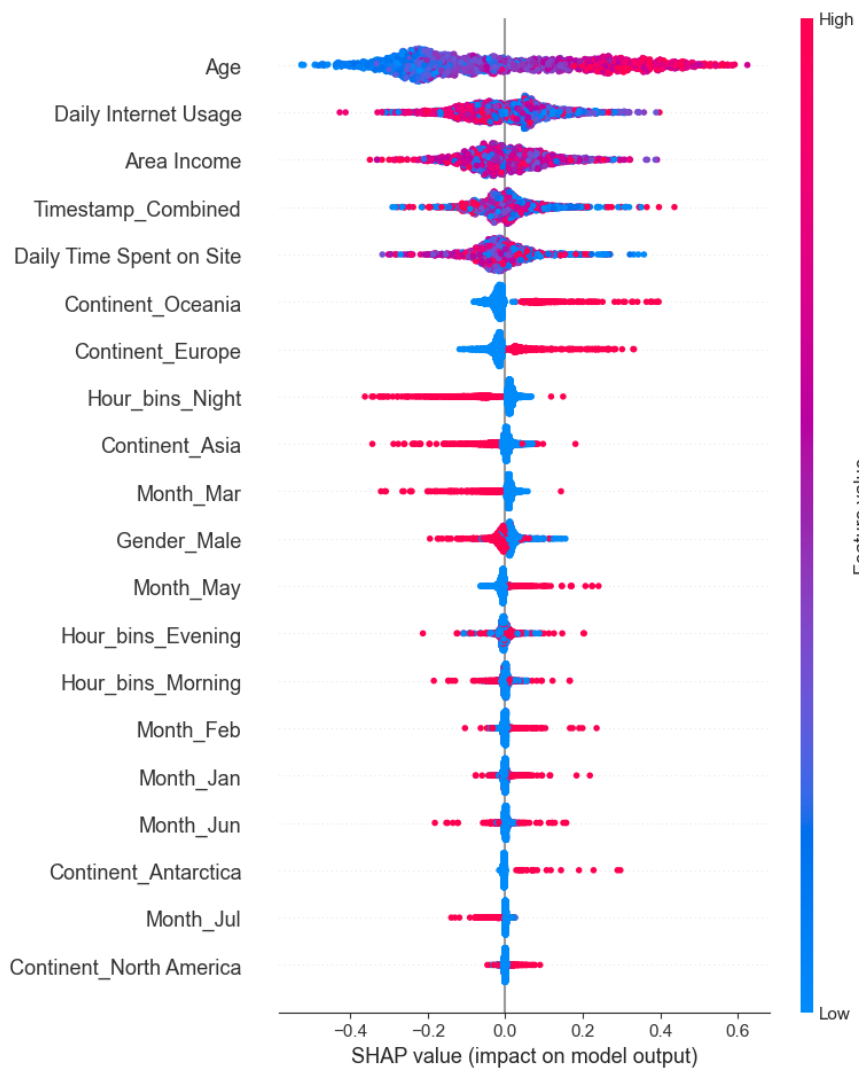
XGBoost

- Train Cross-Val Accuracy: **0.81**
 - Test Accuracy: **0.78**
 - Test ROC_AUC: **0.86**
 - Test F1 Score: **0.78**
- n_estimators: 160
 - max_depth: 8



TREE-BASED MODELS

XGBoost





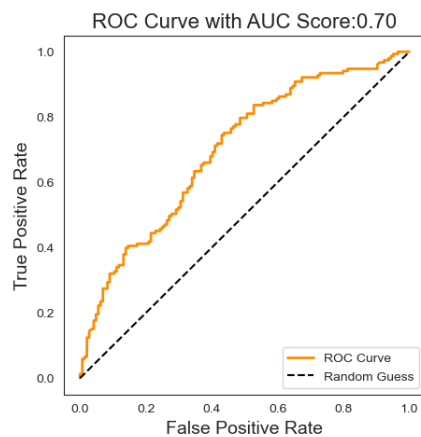
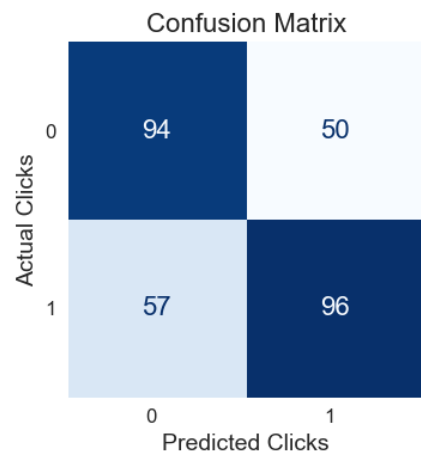
MODEL EVALUATION

(UNSEEN DATA)



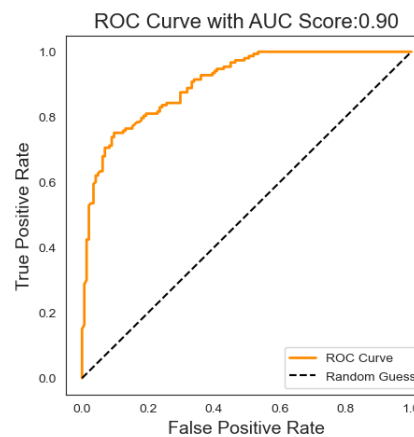
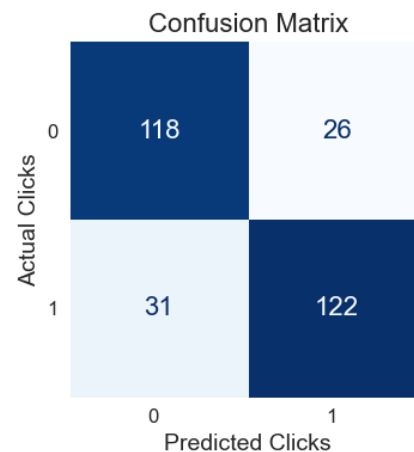
Logistic Regression (Baseline)

Accuracy: **0.63**
ROC_AUC: **0.70**
F1 Score: **0.64**



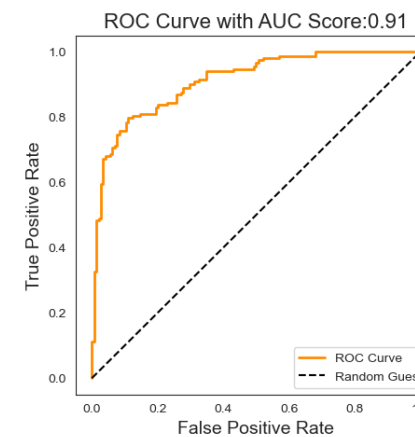
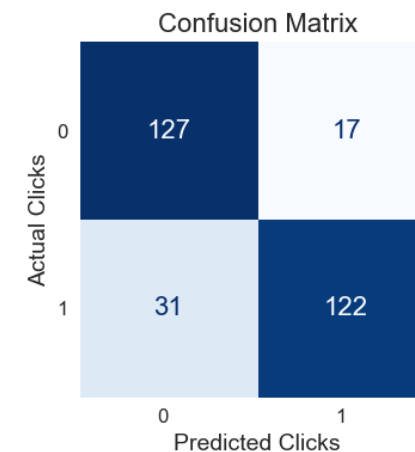
Random Forest (Tuned)

Accuracy: **0.80**
ROC_AUC: **0.90**
F1 Score: **0.81**



XGBoost (Tuned)

Accuracy: **0.83**
ROC_AUC: **0.91**
F1 Score: **0.83**





BUSINESS RECOMMENDATIONS

MARKETING STRATEGIES

- **Target Older Audiences**

Develop campaigns and messaging specifically targeted toward individuals in older age groups.

- **Focus on High Daily Internet Usage**

Target these users through behavior-based segmentation and frequently visited platforms.

- **Optimize Ad Timing**

Deliver ads during afternoon and evening hours when users are more likely to engage based on activity trends.

- **Prioritize High-Income Regions**

Higher-income users show greater ad engagement. Target premium regions and promote high-value or luxury products.

- **Customize Campaigns by Regions**

Analyze and tailor ad strategies for different regions (e.g., Oceania & Europe").



THANK
YOU

Li Wu
Instructor: Reza Moosavi